

Exercise 2.

Question 1

SELECT DISTINCT department
FROM students;

department
IT
HR
IT
Finance
HR.

Question 2.

Get the average age of student per department.

SELECT department, AVG^(age) AS avg_age
FROM students
GROUP BY department;

department	avg-age
IT	20.5
HR	22
Finance	23

Question 3.

Show departments with more than 1 student

SELECT department, COUNT(*) AS student count, AVG(age) AS
FROM students avg_age.
GROUP BY department
HAVING COUNT(*) > 1;

department	avg-age	student count
IT	20,5	2
HR	22	2

Question 4 - Get all students whose age is between 21 and 23

```
SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;
```

Student_id	name	age
2	Bob	22
3	Charlie	21
4	Diana	23
5	Eve	22

Question 5 - List all the IT or HR department who are older than 21.

```
SELECT student_id, name, age, department
FROM students
WHERE department IN ('IT', 'HR')
AND age > 21;
```

Student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR
3	Charlie		

Question 6 - Show total credits per department with more than 5 total credits.

```
SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;
```

department	total_credits
IT	10

Question 7 - List all courses that do not have 4 credits

```
SELECT course_id, course_name, department, credits
FROM courses
WHERE credits <> 4;
```

Course_id	Course_name	department	Credit
101	SQL Basics	IT	3
105	Statistics	HR	3
104	Excel	Finance	2

Question 8 - Show the top 3 courses by credits in descending order

```
SELECT course_id, course_name, credits
FROM courses
ORDER BY credit DESC
LIMIT 3;
```

Course_id	Course_name	department	Credits
102	Python	IT	4
103	Data Science	IT	4
105	Statistics	HR	3

Question 9 - Show the top 3 courses by credits in descending order

```
SELECT course_id, course_name, credits
FROM courses
```


Question 9 - Get the maximum, minimum, and average grade across all enrollments.

```
SELECT MAX(grade) AS max_grade,
       MIN(grade) AS min_grade,
       AVG(grade) AS avg_grade,
FROM enrollments.
```

Max grade	min grade	avg-grade
90	78	$85 + 78 + 90 / 3 = 84,33$

Question 10 - Count how many enrollments exist per course.

```
SELECT course_id,
       COUNT(*) AS total_enrollments
FROM enrollments
GROUP BY course_id;
```

Course_id	total_enrollments
101	1
102	1
103	1
104	1
105	1

Question 11 - Find total salary and total bonus per department

```
SELECT department
      SUM(salary) AS total_salary
      SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department.
```

department	total_salary	total_bonus
IT	122000	11000
HR	109000	9000
Finance	70000	4000

Question 12 - Show departments where average salary is above 55,000

```
SELECT department
      AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;
```

department	avg_salary
IT	62000
Finance	70000

Question 13 - List employees whose salary plus bonus is greater than 60,000.

```
SELECT employee_id, name, salary, bonus, total_compensation
FROM salaries
WHERE (salary + bonus) > 60000.
```

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

Question 14 - Show total and average budget per department
Only include departments with average budget
above 70,000.

```
SELECT department,
       SUM(budget) AS total_budget,
       AVG(budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG(budget) > 70000;
```

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

Question 15 - List all projects with budgets between 50 000
and 120,000, excluding the marketing department.

```
SELECT project_id,
       project_name,
       department,
       budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000
AND department <> 'Marketing';
```

project_id	Project name	department	budget
1	AI APP	IT	120 000
2	Payroll System	Finance	80 000
4	Website Dashboard	Marketing	60 000